Chapter 15, Problem 50P

Problem

Use the Laplace transform to solve the differential equation

$$\frac{d^2v(t)}{dt^2} + 2\frac{dv(t)}{dt} + 10v(t) = 3\cos 2t$$

subject to $v(0) = 1$, $dv(0)/dt = -2$.

Step-by-step solution

Step 1 of 9

The differential equation is,

$$\frac{d^2u(t)}{dt^2} + \frac{2du(t)}{dt} + 10v(t) = 3\cos 2t$$

Apply Laplace transform on both sides.

$$\left[S^2V(S) - SV(0) - V'(0)\right] + 2\left[SV(S) - V(0)\right] + 10V(S) = \frac{3S}{S^2+2^2}$$

$$V(0) = 1$$

And,

$$\frac{dV(0)}{dt} = -2$$
$$V'(0) = -2$$

Step 2 of 9

Hence, the expression is further simplified as,

$$\left[S^2V(S) - S(1) - (-2)\right] + 2\left[SV(S) - 1\right] + 10V(S) = \frac{3S}{S^2+2^2}$$

$$V(S)\left[S^2 + 2S + 10\right] - S + 2 - 2 = \frac{3S}{S^2+4}$$

$$V(S)\left[S^2 + 2S + 10\right] = \frac{3S}{S^2+4} + S$$

$$V(S) = \underbrace{\frac{3S}{(S^2+4)(S^2+2S+10)}}_{V_1(S)} + \underbrace{\frac{S}{(S^2+2S+10)}}_{V_2(S)}$$

Step 3 of 9

Consider the expression $V_1(S)$ as,

$$V_1(S)=\frac{As+B}{S^2+4}+\frac{Cs+D}{S^2+2S+10}$$

Find the values of *A*, *B*, *C*, and *D*.

$$3S=A\left(S^3+2S^2+10S\right)+B\left(S^2+2S+10\right)+C\left(S^3+4S\right)+D\left(S^2+4\right)$$

$$3S=S^3\left(A+C\right)+S^2\left(2A+B+D\right)+S\left(10A+2B+4C\right)+10B+4D \quad \text{...... (1)}$$

Comparing S^3 terms on both sides,

$$A+C=0$$

$$C=-A$$

Comparing S^2 terms on both sides of equation (1),

$$2A+B+D=0 \quad \text{...... (2)}$$

Step 4 of 9

Comparing S terms on both sides of equation (1),

$$10A+2B+4C=B$$

Substitute $-A$ for *C*.

$$10A+2B+4C=B$$

$$6A+2B=3 \quad \text{...... (3)}$$

Comparing constant terms on both sides of equation (1),

$$10B+4D=0$$

$$5B=-2D$$

$$D=-\frac{5}{2}B$$

Substitute $-\frac{5}{2}B$ for *D* in equation (2).

$$2A+B-\frac{5}{2}B=0$$

$$4A-3B=0$$

$$4A=3B$$

$$B=\frac{4}{3}A$$

Step 5 of 9

Substitute $\frac{4}{3}A$ for B in equation (3).

$$6A+2\left(\frac{4}{3}A\right)=3$$

$$18A+8A=9$$

The value of C is,

$$A=\frac{9}{26}$$

$$C=-\frac{9}{26}$$

The value of B is,

$$\begin{aligned}B&=\frac{4}{3}\left(\frac{9}{26}\right)\\&=\frac{6}{13}\end{aligned}$$

The value of D is,

$$\begin{aligned}D&=-\frac{5}{2}\left(\frac{6}{13}\right)\\&=-\frac{15}{13}\end{aligned}$$

Step 6 of 9

Hence the expression for $V_1(S)$ is,

$$\begin{aligned}V_1(S)&=\frac{\frac{9}{26}S+\frac{6}{13}}{S^2+4}+\frac{\frac{-9}{26}S+\left(\frac{-15}{13}\right)}{S^2+2S+10}\\&=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{13}\left(\frac{1}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+\frac{30}{9}}{(S+1)^2+3^2}\right)\\&=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{13}\left(\frac{1}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+1-1+\frac{10}{3}}{(S+1)^2+3^2}\right)\\&=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{13}\left(\frac{1}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+1+\frac{7}{3}}{(S+1)^2+3^2}\right)\end{aligned}$$

Step 7 of 9

Further simplify.

$$V_1(S)=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{13}\left(\frac{1}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+1}{(S+1)^2+3^2}\right)-\frac{9}{26}\left(\frac{\frac{7}{3}}{(S+1)^2+3^2}\right)$$
$$=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{26}\left(\frac{2}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+1}{(S+1)^2+3^2}\right)-\frac{9}{26}\left(\frac{7}{3}\right)\left(\frac{1}{3}\right)\left(\frac{3}{(S+1)^2+3^2}\right)$$
$$=\frac{9}{26}\left(\frac{S}{S^2+4}\right)+\frac{6}{26}\left(\frac{2}{S^2+4}\right)-\frac{9}{26}\left(\frac{S+1}{(S+1)^2+3^2}\right)-\left(\frac{7}{26}\right)\left(\frac{3}{(S+1)^2+3^2}\right)$$

Apply inverse Laplace transform.

$$v_1(t)=\frac{9}{26}\cos 2t+\frac{6}{26}\sin 2t-\frac{9}{26}e^{-t}\cos 3t-\frac{7}{26}e^{-t}\sin 3t$$
$$=\frac{1}{26}\left[9\cos 2t+6\sin 2t-9e^{-t}\cos 3t-7e^{-t}\sin 3t\right]$$

Step 8 of 9

Consider the expression $V_2(S)$ as,

$$V_2(S)=\frac{S}{S^2+2S+10}$$
$$=\frac{S+1-1}{(S+1)^2+3^2}$$
$$=\frac{S+1}{(S+1)^2+3^2}-\frac{1}{(S+1)^2+3^2}$$
$$=\frac{S+1}{(S+1)^2+3^2}-\frac{1}{3}\frac{3}{(S+1)^2+3^2}$$

Step 9 of 9

Apply inverse Laplace transform.

$$v_2(t) = e^{-t}\cos 3t - \frac{1}{3}e^{-t}\sin 3t$$
$$= e^{-t}\left[\cos 3t - \frac{1}{3}\sin 3t\right]$$

The inverse Laplace transform of $V(S)$ is,

$$v(t) = v_1(t) + v_2(t)$$
$$= \frac{1}{26}\left[9\cos 2t + 6\sin 2t - 9e^{-t}\cos 3t - 7e^{-t}\sin 3t\right] + e^{-t}\left[\cos 3t - \frac{1}{3}\sin 3t\right]$$
$$= \frac{1}{26}\left[9\cos 2t + 6\sin 2t\right] + e^{-t}\cos 3t\left[-\frac{9}{26}+1\right] + e^{-t}\sin 3t\left[-\frac{7}{26}-\frac{1}{3}\right]$$
$$= \frac{1}{26}\left[9\cos 2t + 6\sin 2t\right] + e^{-t}\cos 3t\left[-\frac{17}{26}\right] + e^{-t}\sin 3t\left[-\frac{47}{78}\right]$$
$$= \frac{1}{26}\left[9\cos 2t + 6\sin 2t\right] + e^{-t}\left[-17\cos 3t - \frac{47}{3}\sin 3t\right]$$

Therefore, the solution for the differential equation is,

$$\boxed{\frac{1}{26}\left[9\cos 2t + 6\sin 2t\right] + e^{-t}\left[-17\cos 3t - \frac{47}{3}\sin 3t\right]}$$

Comments (1)

Anonymous

should there be a 1/26 for the second part of a outer bracket